Automated Skill Extraction and Resume Matching Using BERT-Based NLP Model

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Project Guide

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Title: Automated Skill Extraction and Resume Matching Using BERT-Based NLP Model

INTRODUCTION

Manual recruitment processes struggle to keep up with the rapid growth of online job listings and applications. Traditional approaches rely heavily on human interpretation and basic keyword matching, which are inefficient and often inaccurate. Our project leverages Natural Language Processing (NLP) and advanced deep learning techniques, particularly BERT (Bidirectional Encoder Representations from Transformers), to automate skill extraction from resumes and job descriptions. By analyzing job market trends, matching candidate profiles to job specifications, and presenting real-time analytics, our system enhances the hiring pipeline's accuracy, transparency, and efficiency.

Technology Used:

- **BERT (Transformer Model):** Captures contextual semantics to improve named entity recognition and text matching.
- **Hugging Face Transformers:** Provides pre-trained NLP models and APIs.
- **Python (NLTK, SpaCy, Scikit-learn):** For text processing and model integration.
- **Neo4j (Graph Database):** For storing and querying interconnected skill-job mappings.
- **PostgreSQL:** For persistent storage of job and resume data.
- **RabbitMQ:** For real-time data ingestion and communication.
- **React.js:** For a dynamic and interactive front-end.

Field of Project:

- **Human Resource Technology:** Enhancing recruitment efficiency and talent mapping.
- **Artificial Intelligence:** Automating NLP tasks such as NER and semantic similarity.
- **Data Analytics:** Extracting insights from job market data for decision-making.

Special Technical Terms:

- **Named Entity Recognition (NER):** Identifies and categorizes skill entities in text.
- **BERTScore:** Measures semantic similarity using contextual embeddings.
- **BIO Tagging:** Labeling scheme for token-level classification.

- **Graph Database:** Stores entity relationships such as skill-job connections.
- **Adversarial Training:** Improves model robustness using synthetic perturbations.

OBJECTIVES

1. Automate the extraction of hard skills, soft skills, and certifications from resumes and job descriptions.
2. Match candidate profiles with job specifications using semantic similarity models.
3. Visualize job market trends across geographies, industries, and experience levels.
4. Improve the accuracy of resume screening using BERT-based models.
5. Implement a real-time job recommendation engine based on extracted skills.
6. Enhance robustness with adversarial inputs and evaluate model performance.

LITERATURE REVIEW

1. Akkasi et al. (2019) - Early Methods in Resume Screening

Akkasi et al. explored the limitations of early methods in resume screening, which predominantly relied on rule-based algorithms and static keyword matching using predefined skill dictionaries. While these approaches were straightforward, they had significant drawbacks, such as the inability to handle synonyms or interpret context. For example, a candidate who describes their skill as "project management" might be overlooked in favor of a candidate who uses the exact phrase "team leadership" if the job description requires the latter. This context-free matching led to the misclassification of many candidates who could be suitable but used different terminology.

In response, advancements were made through the integration of techniques like Conditional Random Fields (CRFs) and Support Vector Machines (SVMs), which allowed models to incorporate some contextual information. However, these models required extensive manual feature engineering, making them difficult to scale and prone to bias. The breakthrough came with the introduction of word embeddings and Long Short-Term Memory networks (LSTMs), which enhanced semantic understanding by capturing relationships between words. However, LSTMs were limited by their unidirectional nature, meaning they could only interpret text in one direction (left to right or right to left), which restricted their ability to understand context effectively.

The advent of transformer-based models like BERT, however, overcame these limitations by providing bidirectional representations, capturing deeper semantic meaning in text. Fine-tuning

BERT on domain-specific data has demonstrated considerable improvements in Named Entity Recognition (NER) tasks, which are crucial for accurately extracting skills and qualifications from resumes and job descriptions.

2. Sun et al. (2020) - Evaluation of Text Similarity

Sun et al. expanded the methods for evaluating text similarity by incorporating both syntactic and semantic distinctions. Early text similarity metrics, such as TER (Translation Edit Rate) and ITER, focused on quantifying the number of edits required to transform one text into another. These methods were useful but often overlooked deeper semantic relationships, focusing instead on surface-level differences.

Subsequent approaches, like PER and CDER, introduced block permutations, which allowed for more nuanced comparisons between texts by considering how blocks of words can be rearranged. Despite these advancements, the need for more sophisticated semantic analysis led to the development of dense word embeddings such as MEANT and YISI-1, which encapsulated more refined semantic similarity.

The most significant development, however, came with the introduction of BERTScore. By leveraging the contextual embeddings from BERT, BERTScore calculates the cosine similarity between tokens from candidate and reference texts. This approach offers a more accurate representation of semantic similarity, as it accounts for word meaning in context rather than just surface-level word overlap. Moreover, BERTScore's incorporation of inverse document frequency (IDF) weights further improves its alignment with human evaluations of text similarity. Despite its promising performance, BERTScore is computationally expensive, which presents a challenge for large-scale applications.

3. Wang et al. (2020) - Adversarial Training in NLP Models

Wang et al. addressed the issue of adversarial examples—inputs that appear normal to humans but cause misclassification by machine learning models. In the context of NLP, adversarial examples are particularly concerning as they can degrade the performance of models in real-world scenarios. TEXTFOOLER, one of the key algorithms discussed in their paper, uses gradient-based techniques to identify crucial words in a text and replace them with synonyms or semantically equivalent terms. This approach generates adversarial examples that retain the overall meaning of the original text but are designed to trick NLP models into making incorrect predictions.

The research demonstrated that training models with adversarial examples could significantly enhance their robustness, as it exposes the models to a broader range of linguistic variations. By incorporating adversarial training, models become more resilient to subtle changes in input data, improving their performance on unseen data and increasing their reliability in real-world applications.

4. Heffernan et al. (2020) - Unsupervised Topic Segmentation of Meeting Transcripts

Heffernan et al. tackled the challenge of segmenting meeting transcripts into coherent topics in the absence of labeled data, an issue commonly encountered in natural language processing tasks. They proposed a novel method utilizing BERT embeddings combined with a semantic similarity scoring technique to segment meeting transcripts. This approach allowed the model to identify logical boundaries between topics without relying on human-annotated data, which is often sparse and difficult to obtain in this domain.

The results showed a 15.5% reduction in error compared to unsupervised baselines and a 26.6% improvement over supervised models trained on non-meeting text. This research highlighted the challenges faced by supervised methods, particularly when working with noisy and unstructured text like meeting transcripts. The authors suggested that incorporating additional signals, such as speaker identification or meeting agendas, could further improve segmentation accuracy and enhance the model's ability to summarize key discussion points effectively.

5. Sun et al. (2020) - Evolution of Text Categorization Methods

Sun et al. discussed the evolution of text categorization in Natural Language Processing (NLP), examining various neural networks that have been employed for classifying text. Early methods, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), were used for modeling text sequences. However, these models struggled with long-range dependencies in text, which limited their effectiveness for tasks like document classification.

Attention mechanisms, particularly in the form of the Transformer architecture, helped address this issue by allowing models to focus on relevant parts of the text regardless of their position. The introduction of pre-trained word embeddings like word2vec and GloVe revolutionized NLP by providing deep semantic representations of words, which could be leveraged for downstream tasks.

Contextual embeddings such as CoVe further enhanced representation learning by capturing context-sensitive semantics. The most significant advancement in the field came with the development of pre-trained transformer models like BERT, which employed a bidirectional design to understand text in both directions, yielding superior results across various NLP benchmarks. Despite its success, the authors noted that further research is needed to refine BERT's fine-tuning techniques for specific tasks like text classification.

6. Bilal et al. (2021) - Machine Learning in Online Review Forecasting

Bilal et al. examined the use of machine learning models in predicting the helpfulness of online reviews. Traditional models, such as Linear Regression and Support Vector Machines (SVMs), relied heavily on manually constructed features like review length, rating, readability, and polarity. While these features provided useful insights, they limited the generalization capacity of the models, as they often overlooked more nuanced aspects of the text.

Recent advances in deep learning, particularly with BERT, have allowed for more sophisticated analysis of unstructured text. BERT-based models eliminate the need for manual feature extraction, instead learning from raw text and capturing deeper semantic patterns. However, Bilal et al. also highlighted that challenges remain in optimizing sequence lengths and selecting appropriate datasets for training, as discrepancies in review characteristics can affect model performance.

7. Thomas et al. (2019) - Web Scraping Techniques

Thomas et al. explored web scraping, a technique used to collect data from websites, which is essential for research, competitive analysis, and trend monitoring. Web scraping enables the extraction of unstructured data from HTML or XHTML pages, and tools like Scrapy, an open-source Python framework, are often employed to facilitate data collection and analysis.

Scrapy's speed and flexibility make it particularly well-suited for handling large-scale data extraction tasks, as it supports asynchronous requests and integrates well with CSS and XPath selectors. The authors also discussed how platforms like Amazon AWS and Google Cloud offer cloud-based solutions for web scraping, making it easier for businesses and researchers to collect and process data at scale. The research emphasized the importance of web structure analysis and automated query processing for improving the efficiency of data extraction.

8. Lotfi et al. (2020) - Web Scraping for Business Intelligence

Lotfi et al. examined the significance of web scraping for business intelligence, marketing analysis, and research across multiple sectors. The authors focused on Scrapy as one of the leading tools for efficient and fast data extraction, highlighting its asynchronous request handling and support for CSS and XPath selectors. These features allow Scrapy to extract large volumes of data quickly, which is crucial for industries that rely on timely insights, such as e-commerce, market research, and financial analysis.

Additionally, services like ScraperAPI improve the efficiency of web scraping by managing proxies and handling CAPTCHAs, which are often obstacles in large-scale data extraction efforts. The paper underscored how web scraping technologies can enable organizations to make informed, data-driven decisions based on real-time information from the web.

9. Xie et al. (2019) - AI and Economic Models

Xie et al. discussed the limitations of traditional economic models, such as the Cobb-Douglas production function, in explaining modern business behavior. These models often fail to account for the internal decision-making processes within organizations, especially in light of advancements in artificial intelligence (AI) and machine learning. The authors presented the Skill-Task Matching Model, which integrates AI algorithms and neural networks to better understand business operations and decision-making.

This model uses vectorized representations of activities, abilities, and business strategies, allowing for more dynamic and iterative learning. It highlights the potential of AI to reconcile the disparity between anticipated and actual profitability by providing insights into the effectiveness of different skill sets in achieving business goals.

10. Chen et al. (2020) - Automated Skill Extraction for Recruitment

Chen et al. focused on the growing significance of automated skill extraction in recruitment and labor market analysis. They highlighted how BERT-based models have shown great promise in extracting skills from unstructured text, such as resumes and job descriptions, due to their ability to understand contextual relationships in language. The Deep BERT-based Semantic Skill Matching (DBSSM) classifier integrates domain-specific modifications and the ESCO taxonomy to enhance precision in skill extraction.

The research demonstrated the effectiveness of these methods by achieving high performance on metrics like MAP@1 and AF1, outperforming traditional approaches. These innovations not only improve candidate shortlisting processes but also provide valuable insights for job seekers, educators, employers, and policymakers in navigating the increasingly complex job market.

FEASIBILITY:

- **Technical Feasibility:** Based on tested machine learning libraries and transformer models, the project is technically viable and scalable.
- **Financial Feasibility:** Requires cloud resources, most of which are freely available under academic credits.
- **Legal Feasibility:** Uses publicly available and anonymized datasets, ensuring compliance with data privacy laws.
- **Operational Feasibility:** Designed to be user-friendly with a simple front-end and backend integration.
- **Security Feasibility:** Ensures data security with anonymization and encrypted communication protocols.

METHODOLOGY:

1. **Data Collection:** Aggregated from Kaggle and scraped LinkedIn job listings using a custom Python scraper.
2. **Data Annotation:** Automated BIO labeling for skill recognition using rule-based NER.
3. **Model Fine-Tuning:** BERT base model fine-tuned on labeled dataset using token classification.
4. **Skill Extraction Pipeline:** Utilizes a token classification head and cross-entropy loss with SoftMax outputs.
5. **Semantic Matching Module:** Employs cosine similarity over sentence embeddings for resume-job alignment.
6. **Trend Analysis:** Neo4j graph database queries visualize trends across role, salary, and geography.
7. **Deployment:** Built using React frontend, Postgres, and RabbitMQ backend with cloud-hosted inference API.

FACILITIES REQUIRED:

- GPU-enabled machine or cloud infrastructure (Google Colab Pro / AWS EC2)
- Python environment with NLP libraries
- Graph DB (Neo4j), SQL database (PostgreSQL)
- Front-end development tools (React.js)
- Message Queue (RabbitMQ)
- Secure data storage and web server for hosting

SIGNIFICANCE:

- **Reduces manual screening time and bias**
- **Improves candidate-job fit and hiring accuracy**
- **Reveals emerging skill trends across industries**
- **Supports workforce planning and upskilling strategies**
- **Provides scalable infrastructure for real-world deployment**

EXPECTED OUTCOME:

A fully functional skill extraction and matching system capable of:

- Extracting skills and certifications from resumes
- Matching candidate profiles to suitable jobs
- Visualizing skill demand across time, role, and geography
- Achieving high accuracy, precision, and recall in NER tasks
- Operating in real-time through an interactive web interface

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FUTURE SCOPE:

- Support for multilingual resumes and job descriptions
- Integration with major ATS systems for enterprise deployment
- Personalized upskilling suggestions based on skill gap analysis
- Incorporation of explainability using SHAP or LIME for model transparency
- Real-time trend monitoring with streaming data pipelines

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